Nisbat va kasr

Sinfda 12 nafar o'g'il bola va 15 nafar qiz bor. Buni nisbat bilan ham aytish mumkin: o'g'il bolalar : qizlar $= 12 : 15 = 4 : 5$, kasr bilan ham: «o'g'il bolalar soni qizlar sonining $\frac{4}{5}$ qismi».

1-QISM

Mavzu



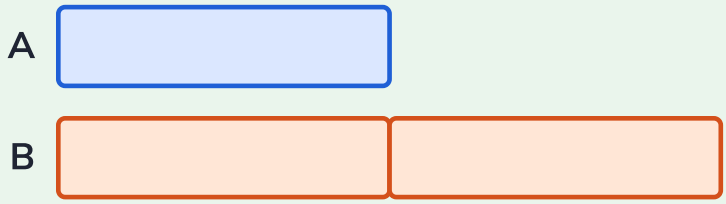
Qaysi miqdor birinchi kelishini adashtirmang!

ESLAB OLAYLIK

Nisbat miqdorlarni taqqoslaydi. 4 ta nok va 11 ta shaftoli: nok : shaftoli $= 4 : 11$, nok : jami $= 4 : 15$. Qisqarmas ko'rinish: $18 : 24 = 3 : 4$ (6 ga bo'ldik).

1-QOIDA · NISBAT VA KASR

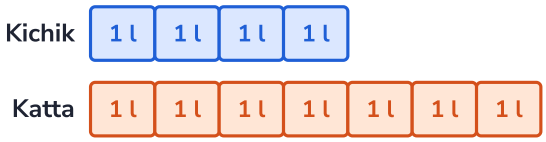
Chizma-modelga ko'ra, $\frac{1}{2}$ va $1 : 2$ yozuvlar bir xil ma'noni anglatadi.



$A : B = 1 : 2$ — A miqdor B miqdorning $\frac{1}{2}$ qismiga teng

1-MISOL · NISBATDAN KASRGA

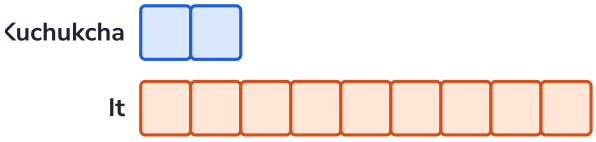
Kichik idishga 4 l, katta idishga 7 l suv sig'adi. Kichik idish katta idishning qanday qismi?



1. Kichik : katta $= 4 : 7$.
2. Kichik idish katta idishning $\frac{4}{7}$ qismi.

2-MISOL · KASRDAN NISBATGA

Kuchukchaning massasi itning massasining $\frac{2}{9}$ qismiga teng. Kuchukcha massasining it massasiga nisbati qanday?



1. Surat 2 — kuchukcha 2 bo'lak, maxraj 9 — it 9 bo'lak.
2. Kuchukcha : it $= 2 : 9$.

Agar miqdorlarni teskari tartibda solishtirsam-chi?



2-QOIDA · HADLAR O'RNINI ALMASHTIRISH

$1 : 2$ nisbat hadlarining o'rnini almashtirib, $2 : 1$ nisbatga ega bo'lamiz.

1-MISOL · IDISHLAR

Kichik : katta $= 4 : 7$. Katta idish kichigining qanday qismi?

1. Hadlar o'rnini almashtiramiz: katta : kichik $= 7 : 4$.
2. Katta idish kichigining $\frac{7}{4}$ qismi. Kasr 1 dan katta, chunki katta idishga ko'proq suv sig'adi.

2-MISOL · NECHA MARTA KO'P

Bog'da 24 tup olma va 6 tup nok daraxti bor.

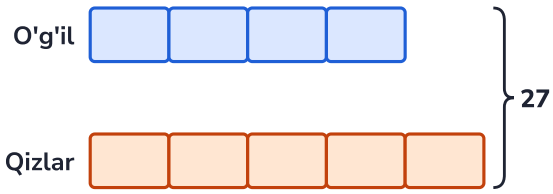
1. Olma : nok $= 24 : 6 = 4 : 1$. $\frac{24}{6} = 4$ — olma nokdan **4 marta** ko'p.
2. Nok : olma $= 1 : 4$ — nok daraxtlari olmaning $\frac{1}{4}$ qismi.

USUL · QISM VA JAMI

Nisbat $2:5$ bo'lsa, jami $2+5=7$ bo'lak. Birinchi miqdor jamining $\frac{2}{7}$, ikkinchisi $\frac{5}{7}$ qismi.

1-MISOL · SINF

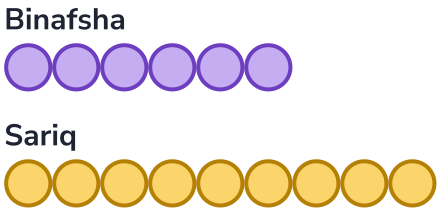
Sinfda 27 nafar o'quvchi bor. O'g'il bolalar : qizlar $= 4:5$.



1. Jami: $4+5=9$ bo'lak. O'g'il bolalar jamining $\frac{4}{9}$ qismi.
2. Jami : o'g'il bolalar $= 9:4$ (hadlar o'rnini almashtirdik).
3. 9 bo'lak = 27, 1 bo'lak = 3. Qizlar: $5 \cdot 3 = 15$ nafar.

2-MISOL · MUNCHOQLAR

Ipga 6 ta binafsha va 9 ta sariq munchoq tizildi. Sariq munchoqlar jamining qanday qismi?



1. Jami: $6+9=15$ ta.
2. $\frac{9}{15} = \frac{3}{5}$; sariq : jami $= 3:5$.

Masalada kasr berilsa, avval kim nechta bo'lak ekanini yozib oling.

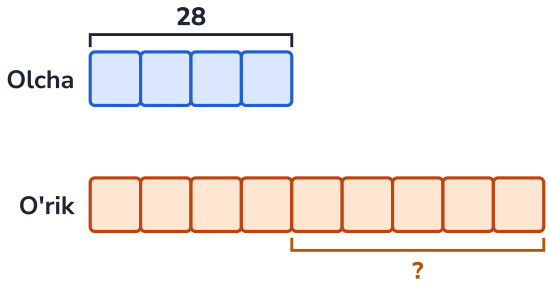


USUL · MATNLI MASALALAR

Kasrdan bo'laklarni yozamiz: $\frac{2}{5}$ bo'lsa — 2 bo'lak va 5 bo'lak. Ma'lum son necha bo'lak ekanini topib, 1 bo'lakni topamiz.

1-MISOL · KO'CHATLAR

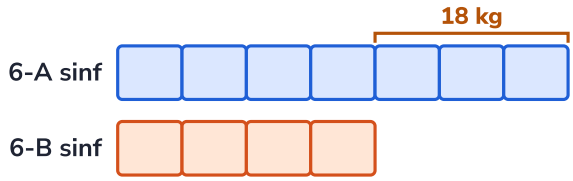
Ko'chatxonada 28 tup olcha ko'chati bor. Bu o'rik ko'chatlarining $\frac{4}{9}$ qismi. O'rik ko'chatlari olchadan necha tup ko'p?



1. Olcha — 4 bo'lak, o'rik — 9 bo'lak. 4 bo'lak = 28, 1 bo'lak = 7.
2. Farq: $9 - 4 = 5$ bo'lak $= 5 \cdot 7 = 35$ tup.

2-MISOL · ESKI QOG'OZ

6-A sinf yig'gan eski qog'oz (makulatura) 6-B sinf yig'ganining $\frac{7}{4}$ qismiga teng va undan 18 kg ko'p. Ikkala sinf jami necha kg yig'gan?



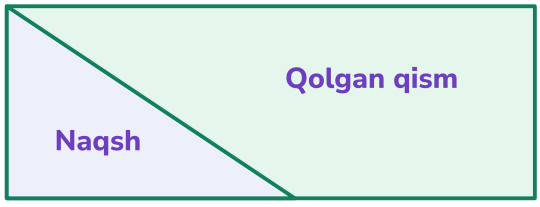
1. 6-A — 7 bo'lak, 6-B — 4 bo'lak. Farq $7 - 4 = 3$ bo'lak = 18 kg, 1 bo'lak = 6 kg.
2. Jami: $11 \cdot 6 = 66$ kg.

2-QISM

Misollar ishlash

1-MISOL · GILAM

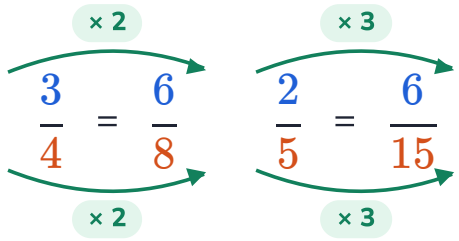
Gilam ichiga uchburchak naqsh tikilgan. Naqsh yuzi qolgan qism yuzining $\frac{3}{8}$ qismiga teng. Naqsh yuzi 30 dm². Butun gilamning yuzi qancha?



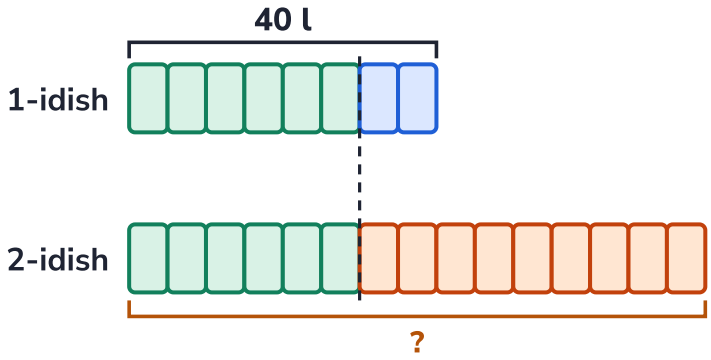
1. Naqsh — 3 bo'lak, qolgan qism — 8 bo'lak.
2. 3 bo'lak = 30 dm², 1 bo'lak $= 30 : 3 = 10$ dm².
3. Butun gilam: $3 + 8 = 11$ bo'lak $= 11 \cdot 10 = 110$ dm². *(Gilam — naqsh va qolgan qism birga.)*

2-MISOL · IDISHLAR

Birinchi idishdagi suvning $\frac{3}{4}$ qismi ikkinchi idishdagi suvning $\frac{2}{5}$ qismiga teng. Birinchi idishda 40 l suv bor. Ikkala idishda jami necha litr suv bor?



1. Suratlarni tenglashtiramiz: $\frac{3}{4} = \frac{6}{8}$, $\frac{2}{5} = \frac{6}{15}$. *(Endi ikkala teng qism 6 bo'lakdan iborat.)*

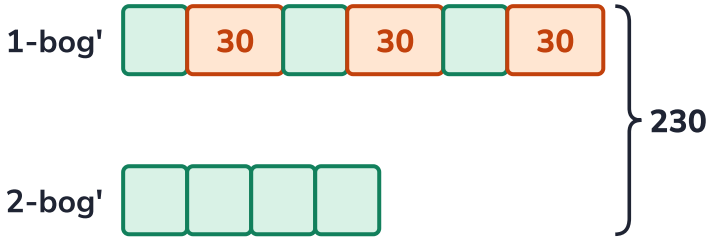


2. 1-idish — 8 bo'lak, 2-idish — 15 bo'lak: nisbat $8 : 15$.
3. 8 bo'lak = 40 l, 1 bo'lak $= 40 : 8 = 5$ l.
4. Jami: $8 + 15 = 23$ bo'lak $= 23 \cdot 5 = 115$ l.
5. Tekshiramiz: 2-idishda 75 l. $\frac{3}{4} \cdot 40 = 30$, $\frac{2}{5} \cdot 75 = 30$ — teng.

Kasr o'rnida butun son bo'lsa: $2 = \frac{2}{1} = \frac{4}{2}$ — suratlarni shunda ham tenglashtiramiz.

3-MISOL · BOG'LAR

Ikki bog'da jami 230 tup daraxt bor. Birinchi bog'dagi daraxtlarning $\frac{1}{3}$ qismi ikkinchi bog'dagilarning $\frac{1}{4}$ qismidan 30 tupga ko'p. Har bir bog'da necha tup daraxt bor?



1. 2-bog'ning $\frac{1}{4}$ qismi — 1 yashil bo'lak, 2-bog' — 4 bo'lak. 1-bog'ning har uchdan biri — 1 yashil bo'lak va yana 30 tup.
2. Ortiqcha: $3 \cdot 30 = 90$ tup. $230 - 90 = 140$. *(Shunda faqat teng bo'laklar qoladi.)*
3. Yashil bo'laklar $3 + 4 = 7$ ta, 1 bo'lak $= 140 : 7 = 20$ tup.
4. 1-bog'ning har uchdan biri $20 + 30 = 50$ tup, uchtasi 150 tup. 2-bog': $4 \cdot 20 = 80$ tup.
5. Tekshiramiz: $150 + 80 = 230$; $150 : 3 = 50$, $80 : 4 = 20$, $50 - 20 = 30$.



DIQQAT!

1. Hadlar tartibi. «Qizlar o'g'il bolalarning $\frac{3}{5}$ qismi»:
   ✗ qizlar : o'g'il bolalar $= 5:3$ → ✓ **qizlar : o'g'il bolalar $= 3:5$**
2. Qism va jami. O'g'il bolalar : qizlar $= 2:3$:
   ✗ o'g'il bolalar jamining $\frac{2}{3}$ qismi → ✓ **jamining $\frac{2}{2+3} = \frac{2}{5}$ qismi**
3. Farq bo'laklarda. Nisbat $7:4$, farq 18:
   ✗ 1 bo'lak $= 18:7$ → ✓ **farq $7-4=3$ bo'lak, 1 bo'lak $= 18:3=6$**
4. Qadoqlarda donalar soni har xil bo'lsa, donalarni sanaymiz. 2 to'rda 5 tadan olma, 3 to'rda 4 tadan nok:
   ✗ olma : nok $= 2:3$ → ✓ $10:12=5:6$
5. Kvadratlarning tomonlari $1:2$ nisbatda bo'lsa:
   ✗ yuzlari ham $1:2$ → ✓ **yuzlari $1:4$ (1 katak va 4 katak)**

3-QISM

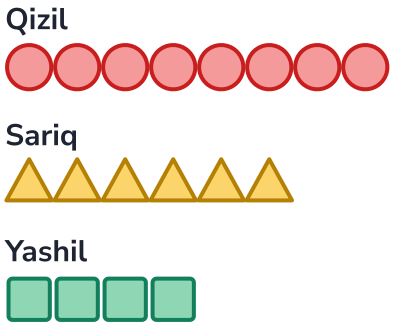
Mustahkamlash

1. Sinfda 12 ta parta va 30 ta stul bor. a) parta : stul; b) partalar stullarning qanday qismi? c) stul : parta; d) stullar partalarning qanday qismi?
2. Gapga mos nisbatni yozing.

Gap	Nisbat
a) Ko'k qalamlar soni qizil qalamlar sonining $\frac{3}{4}$ qismi	ko'k : qizil = ? qizil : ko'k = ?
b) Tarvuz massasi qovun massasining $\frac{5}{2}$ qismi	tarvuz : qovun = ? jami : qovun = ?
c) Umiddagi tangalar Kamrondagilarning $\frac{5}{9}$ qismi	Umid : Kamron = ? Kamron : Umid = ?
d) Tovuqlar o'rdaklardan 7 marta ko'p	tovuq : o'rdak = ? o'rdak : jami = ?

3. Avtoturargohda yengil mashinalar avtobuslardan 6 marta ko'p. a) mashina : avtobus; b) avtobuslar mashinalarning qanday qismi? c) avtobus : jami.

4. Rasmga qarang. a) qizil : sariq nisbatini yozing. b) Yashil shakllar qizillarning qanday qismi? c) Sariq shakllar barcha shakllarning qanday qismi?



5. Javonda 8 ta ertak, 32 ta qissa va 20 ta she'r kitobi bor. a) Qissalar ertaklardan necha marta ko'p? b) She'r kitoblari qissalarning qanday qismi? c) Ertaklar barcha kitoblarning qanday qismi?

6. Rangli qalamlar 12 tadan, oddiy qalamlar 10 tadan qutiga joylangan. Dilnoza 2 quti rangli va 3 quti oddiy qalam oldi. Rangli qalamlar sonining oddiy qalamlar soniga nisbatini qisqarmas ko'rinishda va kasr ko'rinishida yozing.

7. Omborda har 14 ta pomidor qutisiga 6 ta bodring qutisi to'g'ri keladi. a) Pomidor : bodring nisbatini qisqarmas ko'rinishda yozing. b) Bodring qutilari pomidor qutilarining qanday qismi? c) jami : bodring.

8. Kichik akvariumdagi baliqlar katta akvariumdagilarning $\frac{5}{9}$ qismi. a) katta : kichik; b) katta akvariumdagi baliqlar jamining qanday qismi?

9. Velosiped narxi samokat narxining $\frac{9}{4}$ qismiga teng. a) Samokat narxi ikkalasining jami narxining qanday qismi? b) Jami narx velosiped narxining qanday qismi?

10. **Xatoni toping.** «A ning $\frac{2}{3}$ qismi B ning $\frac{3}{4}$ qismiga teng». Dilnoza: «Demak, $A : B = 2 : 3$» dedi. To'g'rimi? To'g'ri nisbatni toping.

11. Ikki kvadratning tomonlari $3 : 4$ nisbatda. Kichik kvadratning yuzi 18 cm². a) Yuzlar nisbati qanday? b) Katta kvadratning yuzini toping.

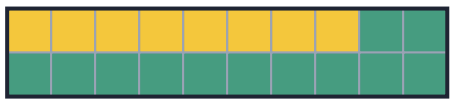
Matnli masalalar

1. Qo'riqxonadagi bug'ular jayronlarning $\frac{5}{8}$ qismini tashkil qiladi. Bug'ular 40 ta. Qo'riqxonada bug'u va jayronlar jami nechta?
2. Katta ko'prikning uzunligi kichik ko'prik uzunligining $\frac{6}{5}$ qismiga teng. Katta ko'prik 156 m. Kichik ko'prik kattasidan necha metr qisqa?
3. Nonvoyxonada bir kunda yopilgan patir nonlar oddiy nonlarning $\frac{5}{12}$ qismi bo'ldi. Oddiy nonlar patirlardan 42 taga ko'p. Nechta patir non yopilgan?
4. To'g'ri to'rtburchakning bo'yi enining $\frac{3}{2}$ qismiga teng. Perimetri 90 cm. To'g'ri to'rtburchakning yuzini toping.
5. Birinchi qopda 48 kg un bor. Birinchi qopdagi unning $\frac{1}{4}$ qismi ikkinchi qopdagi unning $\frac{3}{5}$ qismiga teng. Ikkinchi qopda birinchisidan necha kg kam un bor?

4-QISM

Mustaqil ish

1. Qutidan 8 ta sariq va 12 ta yashil kubik olindi. Kasr orqali ikkita va nisbat orqali ikkita gap yozing.



2. «R ning $\frac{9}{4}$ qismi $= S$». Chizmali model chizing. a) $S : R$ va $R : S$ nisbatlarini yozing. b) S miqdor jami $(S + R)$ ning qanday qismi?
3. Birinchi tokchadagi idishlarning $\frac{4}{7}$ qismi ikkinchi tokchadagi idishlarning 2 baravariga teng. Birinchi tokchada ikkinchisidagidan 45 ta ko'p idish bor. Ikkala tokchada jami nechta idish bor?
4. 4 dona pishiriq uchun: 2 stakan un, $\frac{1}{2}$ stakan sut, $2\frac{1}{2}$ osh qoshiq shakar, 4 dona tuxum kerak. 6 dona pishiriq uchun har bir masalliqdan qancha kerak? *(Avval 1 dona uchun 4 ga bo'ling.)*

5. Ikki javonda jami 124 ta kitob bor. Birinchi javondagi kitoblarning $\frac{1}{2}$ qismi ikkinchi javondagilarning $\frac{1}{3}$ qismidan 12 taga ko'p. Har bir javonda nechta kitob bor?
6. Ikki bochkaga 60 litrdan suv quyildi. Ulardan jami 48 l suv ishlatildi. Shundan keyin birinchi bochkadagi suvning $\frac{2}{5}$ qismi ikkinchi bochkadagi suvning $\frac{1}{2}$ qismiga teng bo'lib qoldi. Har bir bochkadan necha litr suv ishlatilgan?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $12 : 30 = 2 : 5$; b) $\frac{2}{5}$; c) $5 : 2$; d) $\frac{5}{2}$.

2-masala

a) $3 : 4$; $4 : 3$. b) $5 : 2$; jami $5 + 2 = 7$ bo'lak, $7 : 2$. c) $5 : 9$; $9 : 5$. d) $7 : 1$; jami 8 bo'lak, $1 : 8$.

3-masala

a) $6 : 1$; b) $\frac{1}{6}$; c) jami $6 + 1 = 7$ bo'lak, $1 : 7$.

4-masala

a) $8 : 6 = 4 : 3$; b) $\frac{4}{8} = \frac{1}{2}$; c) jami 18 ta, $\frac{6}{18} = \frac{1}{3}$.

5-masala

a) $\frac{32}{8} = 4$ marta; b) $\frac{20}{32} = \frac{5}{8}$; c) jami 60 ta, $\frac{8}{60} = \frac{2}{15}$.

6-masala

Rangli: $2 \cdot 12 = 24$ ta, oddiy: $3 \cdot 10 = 30$ ta. $24 : 30 = 4 : 5$; rangli qalamlar oddiylarning $\frac{4}{5}$ qismi.

7-masala

a) $14 : 6 = 7 : 3$; b) $\frac{3}{7}$; c) jami $7 + 3 = 10$ bo'lak, $10 : 3$.

8-masala

a) $9 : 5$; b) jami $9 + 5 = 14$ bo'lak, $\frac{9}{14}$.

9-masala

Velosiped — 9 bo'lak, samokat — 4 bo'lak, jami 13 bo'lak. a) $\frac{4}{13}$; b) $\frac{13}{9}$.

10-masala

To'g'ri emas. Suratlarni tenglashtiramiz: $\frac{2}{3} = \frac{6}{9}$, $\frac{3}{4} = \frac{6}{8}$. A — 9 bo'lak, B — 8 bo'lak: $A : B = 9 : 8$.

11-masala

a) Kichik kvadrat $3 \cdot 3 = 9$ katak, katta $4 \cdot 4 = 16$ katak: $9 : 16$. b) 9 bo'lak = 18 cm², 1 bo'lak = 2 cm², katta kvadrat $16 \cdot 2 = 32$ cm².

Matnli masalalar

1-masala

Bug'ular — 5 bo'lak = 40, 1 bo'lak = 8. Jami $5 + 8 = 13$ bo'lak $= 13 \cdot 8 = 104$ ta.

2-masala

6 bo'lak = 156 m, 1 bo'lak = 26 m. Farq $6 - 5 = 1$ bo'lak $= 26$ m.

3-masala

Patir — 5 bo'lak, oddiy — 12 bo'lak. Farq 7 bo'lak = 42, 1 bo'lak = 6. Patir: $5 \cdot 6 = 30$ ta.

4-masala

Eni — 2 bo'lak, bo'yi — 3 bo'lak. Bo'yi va eni birga $90 : 2 = 45$ cm, bu 5 bo'lak, 1 bo'lak = 9 cm. Eni 18 cm, bo'yi 27 cm. Yuzi $27 \cdot 18 = 486$ cm².

5-masala

$\frac{1}{4} = \frac{3}{12}$ va $\frac{3}{5}$: 1-qop : 2-qop $= 12 : 5$. 12 bo'lak = 48 kg, 1 bo'lak = 4 kg. Farq $12 - 5 = 7$ bo'lak $= 28$ kg.

Mustaqil ish

1-masala

Masalan: sariq kubiklar yashillarning $\frac{8}{12} = \frac{2}{3}$ qismi; sariq kubiklar jamining $\frac{8}{20} = \frac{2}{5}$ qismi. Yashil : sariq $= 3:2$; yashil : jami $= 3:5$.

2-masala

S — 9 bo'lak, R — 4 bo'lak. a) $S:R = 9:4$, $R:S = 4:9$. b) Jami 13 bo'lak, S jamining $\frac{9}{13}$ qismi.

3-masala

$2 = \frac{2}{1} = \frac{4}{2}$, $\frac{4}{7}$ — suratlar teng. 1-tokcha : 2-tokcha $= 7:2$. Farq 5 bo'lak = 45, 1 bo'lak = 9. Jami $9 \cdot 9 = 81$ ta (63 va 18).

4-masala

Masalliq	4 dona	1 dona (: 4)	6 dona (× 6)
un (stakan)	2	$\frac{1}{2}$	3
sut (stakan)	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{3}{4}$
shakar (osh qoshiq)	$2\frac{1}{2}$	$\frac{5}{8}$	$3\frac{3}{4}$
tuxum (dona)	4	1	6

5-masala

2-javon — 3 bo'lak; 1-javonning har yarmi — 1 bo'lak va yana 12. Ortiqcha: $2 \cdot 12 = 24$. $124 - 24 = 100$ — bu $2 + 3 = 5$ bo'lak, 1 bo'lak = 20. 1-javon: $2 \cdot 20 + 24 = 64$ ta, 2-javon: $3 \cdot 20 = 60$ ta.

6-masala

Qolgan jami: $120 - 48 = 72$ l. $\frac{1}{2} = \frac{2}{4}$ va $\frac{2}{5}$: qolganlar $5:4$. 9 bo'lak = 72 l, 1 bo'lak = 8 l. Birinchisida 40 l, ikkinchisida 32 l qoldi. Ishlatilgan: $60 - 40 = 20$ l va $60 - 32 = 28$ l.